

✓ DECO6500 · A2 Quantitative First Pass

Version 1.1 · September 2026 · DECO6500 Advanced Human-Computer Interaction, UQ

⚠ Before you do anything else: **File → Save a copy in Drive**

You are looking at the course's master copy. You cannot save changes to it, and anything you type here disappears when the session ends. Save your own copy first (or click **Copy to Drive** in the toolbar), then work in that copy. Do this every time you open the notebook from the course link.

Course link to this notebook:
https://colab.research.google.com/github/petewnr/DECO6500-A2-Quantitative/blob/main/DECO6500_A2_Quantitative_First_Pass.ipynb

What may go into this notebook

Only **de-identified quantitative data**: participant IDs (P01, P02 ...), demographics and psychographics, task outcomes, times, error counts, questionnaire scores. **Never** names, contact details, recordings, or transcripts. Transcripts go into NVivo through the UQ Digital Workspace. If you are unsure whether a column is identifying, leave it out. The master copy of your data stays in your Team SharePoint folder; this notebook works on an upload that is deleted when the session closes.

What this notebook does

It takes one CSV — one row per participant — and produces the things the Week 7 lecture described: counts before percentages, medians and ranges, dot plots that show every value, SUS and NASA-TLX scoring, a scatterplot with Spearman and Pearson for **association**, and a two-group comparison with Mann-Whitney or Fisher's exact for **difference**. You do not need to write Python. You tell it your column names once (Section 3), edit the short lists marked **EDIT**, and run the cells in order (▶ on the left of each cell, or Shift+Enter).

What it will not do for you

It will not tell you what a number means for your committed questions. Every output here is a description of your sixteen to twenty participants, and the register is **suggests**. Decide *before* you run Sections 8 and 9 which two or three associations or comparisons your committed questions care about, write them in the 'decided in advance' cells, and report those — including the ones that show nothing.

The notes cells are the point

After each analysis step there is a **Analysis notes** cell. Double-click it and write what you see, in sentences, while it is in front of you. Those notes — not the charts — are what goes into your research notebook and the report, and they are what a tutor will read on Friday. A notebook full of outputs and empty notes cells is a notebook where no analysis has happened.

> 1 · Setup — run this cell once

Loads the libraries and defines the helper functions the rest of the notebook uses. Nothing to edit.

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> 2 · Load your data — run **one** of the two cells below

2A loads the built-in sample (the study planner from the Week 6 clinic: 18 participants, 6 interviewed). Run it first, the first time, so you can see what every section produces.

2B uploads your team's own de-identified CSV. Run it instead of 2A when you are ready to work on your data. If you run both, whichever ran **last** is the data the rest of the notebook uses.

What your CSV needs to look like

One row per participant. Column names are up to you — but **no spaces** (use `_`), and you will tell the notebook what they are in Section 3. The sample uses these, and copying its layout is the easiest route:

Column	What it holds
<code>participant_id</code>	P01, P02 ... — never a name
<code>age_band</code> , <code>prior_planning_habit</code> , <code>confidence_tech</code> ...	the demographics and psychographics you recorded for everyone

Column	What it holds
<code>interviewed</code>	<code>(yes)</code> / <code>(no)</code> — the six who also did the qualitative part
<code>task1_outcome</code> ...	<code>unaided</code> / <code>assisted</code> / <code>not_completed</code>
<code>task1_time_s</code> ...	seconds, for completers; leave blank if not completed
<code>task1_errors</code> ...	a count, using the error definition you wrote down before counting
<code>sus_1</code> ... <code>sus_10</code>	the ten SUS item responses (1–5), if you used SUS
<code>tlx_mental</code> ... (six columns starting <code>tlx_</code>)	NASA-TLX sub-scales 0–100, if you used it
<code>control_item</code> ...	any single Likert item, 1–5

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3 · Tell the notebook about your columns, check the data, and describe the people

 **EDIT the cell below** so every name matches a column in *your* CSV. If you are using the sample, leave it as is. Set anything you did not collect to `(None)` or an empty list.

The check cell after it tells you which names it cannot find, and shows the column types: if a time column shows as `(object)` rather than a number, there is text in it somewhere (a stray "n/a", a comma) — fix the CSV, not the notebook.

The third cell describes **the people** — a count at each value of every person-level variable you recorded (the demographics and psychographics), for the whole group and for the interviewed participants beside it. This is the "who took part" paragraph of your report, and it is the first look at whether the interviewed six resemble the rest (Section 10 checks that against the outcomes).

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4 · Counts — the outcome of each task

Counts with their n, and the percentage **beside** the count, never instead of it. Unaided and assisted are kept apart: a task completed after the facilitator helped is not the same evidence as one completed alone.

Report as: *"Fifteen of eighteen participants (83%) completed Task 1 unaided; two completed it with assistance; one did not complete it."*

```
ORDER = [COLS["labels"]["unaided"], COLS["labels"]["assisted"], COLS["labels"]["not_completed"]]
for col in COLS["outcomes"]:
    print(f"— {col} —")
    display(count_table(df, col, order=ORDER))
```

```
— Overall_Task_Outcome —
```

	count	of_n	percent	report_as
Overall_Task_Outcome				
unaided	15	16	94	15 of 16 (94%)
assisted	1	16	6	1 of 16 (6%)

Analysis notes — Section 4: Overall Task Outcome

Fifteen of sixteen participants (94%) completed the task unaided, while one participant (6%) completed it with assistance. No participant was classified as `(not_completed)`, and all participants reached a viable final travel action. This bears directly on the Layer 1.1 evaluation question of whether participants can understand the disruption situation and use the prototype to make an appropriate next travel decision without assistance.

The register is **suggests**: the large majority of participants were able to complete the task independently, which suggests that the prototype generally supported independent decision-making in this sample. However, the result does not establish that all international students would be able to do the same, particularly because the sample contained only moderate- and high-frequency public transport users.

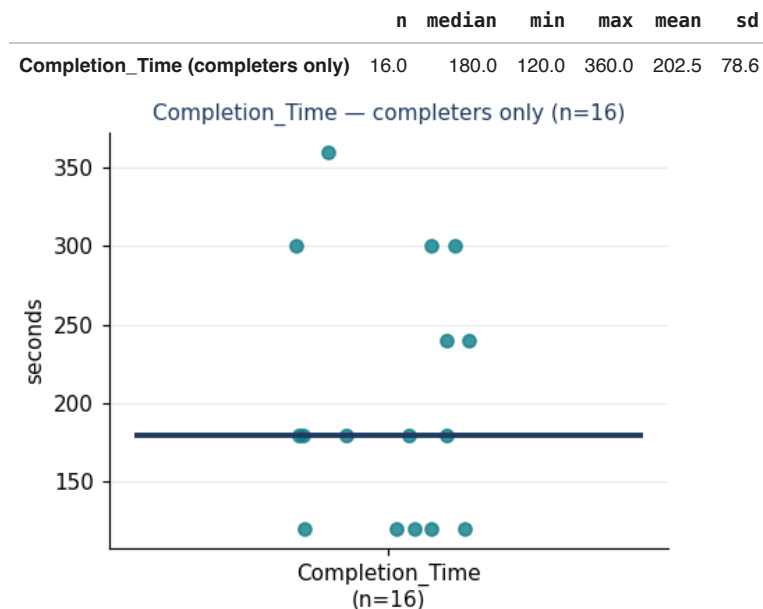
The assisted completion should be examined in the qualitative and observational data to identify where support was required and whether the difficulty came from interpreting the disruption, understanding its impact, comparing alternatives, or selecting the final action. The qualitative data should also be checked for participants who were classified as unaided but still showed hesitation or uncertainty, because successful task completion does not necessarily mean that the process was effortless or fully clear.

The classification used here should also be kept in mind when interpreting the result: assistance levels 0–1 were treated as `unaided`, while level 2 or above was treated as `assisted`. This means the outcome reflects independent completion under the study's predefined assistance criteria rather than the complete absence of any moderator interaction.

5 · Summaries and dot plots — times and errors

Median and range for times (one person who got stuck drags the mean towards them). Times are for **completers only** — a non-completion is not a time. The dot plot shows every participant, with the median as a bar; with eighteen people that is more honest than any summary alone.

```
COMPLETED = [COLS["labels"]["unaided"], COLS["labels"]["assisted"]]  
  
def completers(df, time_col):  
    """Rows where the task paired with this time column was completed (unaided or assisted)."""  
    gate = COLS["times"].get(time_col)  
    return df[df[gate].isin(COMPLETED)] if gate else df  
  
rows = [summarise(completers(df, col), col, label=f"{col} (completers only)") for col in COLS["times"]]  
rows += [summarise(df, col) for col in COLS["errors"]]  
display(pd.DataFrame(rows))  
  
for col in COLS["times"]:  
    sub = completers(df, col)  
    dot_plot(sub, col, title=f"{col} — completers only (n={len(sub)})", ylabel="seconds")
```



Analysis notes — Section 5: Completion Time

What I see:

All 16 participants completed the task. The median completion time was 180 seconds, with a range from 120 to 360 seconds. The mean completion time was 202.5 seconds (SD = 78.6 seconds). Most participants completed the task between 120 and 300 seconds, while one participant took 360 seconds.

Which committed question this bears on:

This bears on the Layer 1.1 usability question by describing how efficiently participants were able to work through the prototype and reach a final travel decision.

Register — demonstrates / suggests / cannot establish — and why:

This **demonstrates** the completion-time distribution for these 16 participants. It **suggests** that most participants were able to complete the task within a relatively short period, but the spread in completion times indicates that some participants required substantially more time than others. It cannot establish that the prototype would produce the same completion times for international students more generally.

What the interviews need to explain, or what I want to check next:

The qualitative data should be checked to understand why some participants took longer than others, especially the participant

who required 360 seconds. Relevant explanations may include hesitation, difficulty interpreting disruption information, comparing alternatives, uncertainty about the final decision, or differences in prior public transport experience.

Deviations from the plan that affect this:

No major procedural deviation directly affects the completion-time measure. However, because the sample includes only moderate- and high-frequency public transport users, completion times may partly reflect participants' prior familiarity with public transport rather than the prototype alone.

> 6 · Standardised instruments — SUS and NASA-TLX

Runs whichever you set in `COLS`. **SUS**: report the mean, SD, range and n. 68 is the commonly cited midpoint across many published studies, not a pass mark, and with n = 18 the score is indicative. **NASA-TLX**: report the six sub-scales, not only the raw total — the total hides *which kind* of load.

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✓ 7 · Single Likert items

A single 1–5 item is **ordinal**: report the count at each point (or the median), not a mean of the numbers.

```
for col in COLS["likert"]:
    print(f"— {col} —")
    display(count_table(df, col, order=[1, 2, 3, 4, 5]))
    print(f"median = {pd.to_numeric(df[col], errors='coerce').median()}\n")
```


— Rating_Disruption_Clarity —

	count	of_n	percent	report_as
Rating_Disruption_Clarity				
2	1	16	6	1 of 16 (6%)
3	2	16	12	2 of 16 (12%)
4	5	16	31	5 of 16 (31%)
5	8	16	50	8 of 16 (50%)

median = 4.5

— Rating_Compare_Options —

	count	of_n	percent	report_as
Rating_Compare_Options				
3	2	16	12	2 of 16 (12%)
4	11	16	69	11 of 16 (69%)
5	3	16	19	3 of 16 (19%)

median = 4.0

— Rating_Next_Step_Clarity —

	count	of_n	percent	report_as
Rating_Next_Step_Clarity				
3	3	16	19	3 of 16 (19%)
4	6	16	38	6 of 16 (38%)
5	7	16	44	7 of 16 (44%)

median = 4.0

— Rating_Decision_Confidence_Info —

	count	of_n	percent	report_as
Rating_Decision_Confidence_Info				
2	1	16	6	1 of 16 (6%)
3	4	16	25	4 of 16 (25%)
4	4	16	25	4 of 16 (25%)
5	7	16	44	7 of 16 (44%)

median = 4.0

— Decision_Confidence —

	count	of_n	percent	report_as
Decision_Confidence				
3	3	16	19	3 of 16 (19%)
4	8	16	50	8 of 16 (50%)
5	5	16	31	5 of 16 (31%)

median = 4.0

— PU1 —

	count	of_n	percent	report_as
PU1				
2	2	16	12	2 of 16 (12%)
3	2	16	12	2 of 16 (12%)
4	6	16	38	6 of 16 (38%)
5	6	16	38	6 of 16 (38%)

median = 4.0

— PU2 —

	count	of_n	percent	report_as
PU2				
2	3	16	19	3 of 16 (19%)
4	5	16	31	5 of 16 (31%)

5 8 16 50 8 of 16 (50%)
median = 4.5

— PU3 —

	count	of_n	percent	report_as
PU3				

2	2	16	12	2 of 16 (12%)
3	2	16	12	2 of 16 (12%)
4	3	16	19	3 of 16 (19%)
5	9	16	56	9 of 16 (56%)

median = 5.0

— PU4 —

	count	of_n	percent	report_as
PU4				

2	1	16	6	1 of 16 (6%)
3	5	16	31	5 of 16 (31%)
4	5	16	31	5 of 16 (31%)
5	5	16	31	5 of 16 (31%)

median = 4.0

— PEOU1 —

	count	of_n	percent	report_as
PEOU1				

2	1	16	6	1 of 16 (6%)
4	9	16	56	9 of 16 (56%)
5	6	16	38	6 of 16 (38%)

median = 4.0

— PEOU2 —

	count	of_n	percent	report_as
PEOU2				

3	3	16	19	3 of 16 (19%)
4	6	16	38	6 of 16 (38%)
5	7	16	44	7 of 16 (44%)

median = 4.0

— PEOU3 —

	count	of_n	percent	report_as
PEOU3				

2	2	16	12	2 of 16 (12%)
3	1	16	6	1 of 16 (6%)
4	8	16	50	8 of 16 (50%)
5	5	16	31	5 of 16 (31%)

median = 4.0

— PEOU4 —

	count	of_n	percent	report_as
PEOU4				

3	3	16	19	3 of 16 (19%)
4	7	16	44	7 of 16 (44%)
5	6	16	38	6 of 16 (38%)

median = 4.0

— Confidence —

	count	of_n	percent	report_as
Confidence				

Analysis notes — Sections 6–7: Questionnaire and Likert Evidence

Information clarity and decision support

Fourteen of sixteen participants rated the ease of comparing travel options as 4 or 5, and thirteen of sixteen rated next-step clarity as 4 or 5; both measures had a median of 4. This bears on the evaluation question of whether the prototype helps participants understand the disruption situation and decide what to do next. The register is **suggests**: most participants rated the decision-support process positively, but the ratings alone cannot show which specific information or interface element made the process clear. The interview data should therefore be used to identify which parts of the information supported comparison and action selection, and where participants still experienced hesitation or confusion.

Perceived usefulness

The strongest usefulness result concerned deciding whether the current travel plan needed to change: 9 of 16 participants selected the highest rating of 5, with a median of 5. Understanding how the disruption affected the participant's own journey was also rated strongly, with 8 of 16 selecting 5 and a median of 4.5. The broader usefulness measures remained positive, with medians of 4. This bears on the question of whether the prototype provides meaningful decision support during disruption. The register is **suggests**: the prototype appears to be particularly useful for helping participants judge whether action is necessary, rather than simply presenting disruption information. The interviews should explain what information participants relied on when deciding whether to continue their current journey or change plans.

Decision confidence

Thirteen of sixteen participants rated their overall decision confidence as 4 or 5, with a median of 4. However, the ratings specifically concerning whether the prototype information increased confidence were more dispersed, despite also having a median of 4. This bears on the question of whether the prototype reduces uncertainty and supports confidence in travel decisions. The register is **suggests**: participants were generally confident in their final decisions, but the quantitative results do not establish that this confidence came entirely from the prototype. The interviews should therefore examine whether confidence came from the presented information, prior public transport experience, route familiarity, or knowledge from other transport tools.

Perceived ease of use

All four ease-of-use measures had a median of 4, and most responses were concentrated at 4 or 5. This bears on the evaluation question of whether participants found the prototype understandable and easy to use. The register is **suggests**: participants generally perceived the prototype as learnable, understandable, and manageable to navigate, but this sample contained only moderate- and high-frequency public transport users. Some of the perceived ease may therefore reflect prior familiarity with public transport rather than the interface alone. The qualitative data should be checked for any usability difficulties that were not visible in the positive ratings.

SUS and NASA-TLX were not used in this study, so Section 6 did not produce a standardised usability or workload score. The evidence in this section instead comes from the project-specific quick-evaluation items and the adapted TAM measures, and should be interpreted descriptively for this participant sample.

✓ 8 · Association — does a characteristic of the person go with an outcome?

This is where the demographics and psychographics you recorded for everyone earn their place. **Before running anything, write in the notes cell below which two or three associations your committed questions care about, and why.** Then run those, report all of them (including the ones that show nothing), and mark anything else as exploratory. Six person-variables against five outcomes is thirty correlations; one or two will look interesting by chance.

The cell draws the **scatterplot first**, then gives **Spearman p** (the default — Likert items are ordinal and small samples are rarely normal) and Pearson r, each with n and p. Read the size, not just the sign, and read the plot before the coefficient: at this n one dot can make the line.

Decided in advance — Section 8 (association)

Formal Section 8 associations:

No formal association was run because the available participant-level background variable, public transport use frequency, is categorical (**Moderate** / **High**) and is more appropriate for the two-group comparison in Section 9.

Exploratory association:

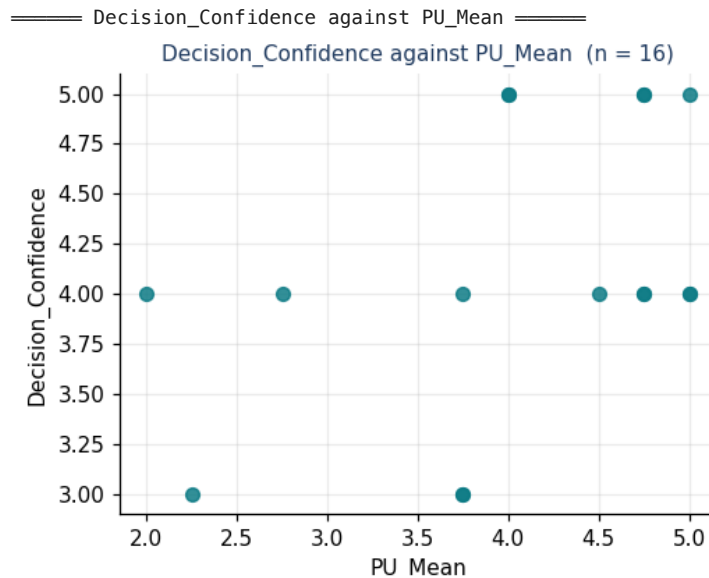
Perceived usefulness against decision confidence — explored to see whether participants who perceived the prototype as more

useful also tended to report higher decision confidence. This was not pre-specified as a formal Section 8 association because both variables are outcome measures rather than participant characteristics.

Anything I run beyond these is **exploratory** and will be labelled as such.

```
# EDIT: each pair is (person variable, outcome). These should match what you wrote in the cell above.
# Exploratory analysis – not pre-specified
ASSOCIATIONS = [
    ("PU_Mean", "Decision_Confidence"),
]

for x, y in ASSOCIATIONS:
    print(f"===== {y} against {x} =====")
    d = completers(df, y) if y in COLS["times"] else df      # times: completers only
    associate(d, x, y)
    print()
```



Spearman $\rho = 0.46$, $p = 0.071$, $n = 16$ ← report this one by default
Pearson $r = 0.44$, $p = 0.086$, $n = 16$ ← only for two continuous, roughly normal measures

Read: a moderate positive association in these 16 participants. Would not conventionally be called statistically significant. That is a reportable result, not a failure: with $n = 16$ a correlation needs to be about 0.50 or larger before $p < .05$. Report ρ , n and p together. Association is not cause. Write the sentence in the 'suggests' register and point to the interviews for why.

Analysis notes — Section 8 (association)

Exploratory association — perceived usefulness and decision confidence

The scatterplot shows a moderate positive pattern between perceived usefulness and decision confidence. Spearman's ρ was 0.46 ($n = 16$, $p = .071$). This does not reach the conventional $p < .05$ threshold, but it suggests that participants who perceived the prototype as more useful also tended to report higher decision confidence.

Because this analysis was exploratory rather than pre-specified, it should not be treated as a confirmatory finding. The association also does not establish causation. The interview data should be used to examine whether higher confidence was related to the prototype's usefulness, prior public transport experience, route familiarity, or other sources of information.

9 · Difference — do two kinds of participant differ on an outcome?

Two groups only; nine and nine is the most the sample can stand, and a group of two or three is not a group. If you split a Likert item into two groups, decide the split point **before** you look at the outcome (4–5 versus 1–3, say) and write down why.

The cell shows the per-group descriptives and the two-column dot plot first, then the test: **Mann–Whitney U** for a measured outcome (time, SUS, errors) or **Fisher's exact** for a yes/no outcome (completed or not — give the `success_value`). Report the difference, then the test. A null result is "no statistically significant difference" — it is never "no difference" or "the groups performed the same".

Decided in advance — Section 9 (difference)

Comparison 1: Completion time between participants with High and Moderate public transport use frequency — because the evaluation asks whether participants can efficiently interpret disruption information and reach a travel decision, and prior transport familiarity may affect how quickly they complete the task.

Comparison 2: Decision confidence between participants with High and Moderate public transport use frequency — because the evaluation asks whether the prototype supports confident decision-making, and prior public transport familiarity may influence participants' confidence when responding to disruption information.

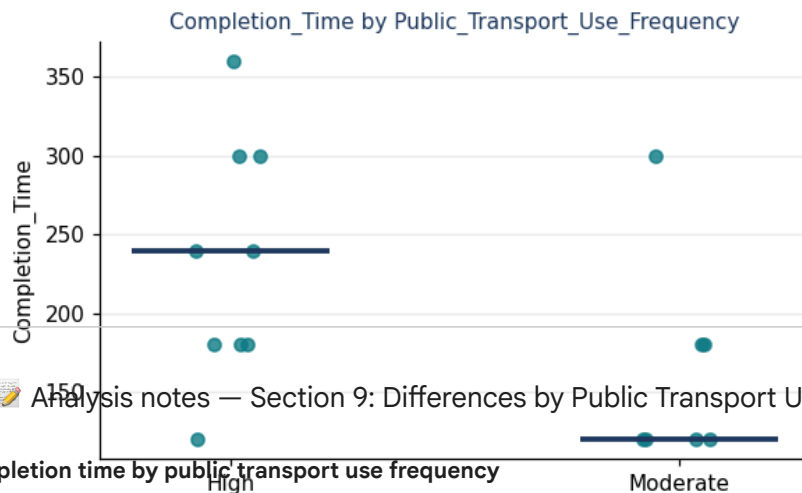
Anything beyond these is **exploratory**.

```
# Example: make a two-level grouping from a Likert item, with the split point stated.  EDIT or delete.
#df["confidence_group"] = np.where(pd.to_numeric(df["confidence_tech"], errors="coerce") >= 4, "high (4-5)"

#  EDIT: each entry is (outcome, grouping column, success_value or None). Match what you wrote above.
COMPARISONS = [
    ("Completion_Time", "Public_Transport_Use_Frequency", None),
    ("Decision_Confidence", "Public_Transport_Use_Frequency", None),
]

for outcome, group, success in COMPARISONS:
    print(f"===== {outcome} by {group} =====")
    d = completers(df, outcome) if outcome in COLS["times"] else df
    compare(d, outcome, group, success_value=success)
    print()
```

Completion_Time by Public_Transport_Use_Frequency				
	n	median	min	max
Public_Transport_Use_Frequency				
High	9	240.0	120	360
Moderate	7	120.0	120	300



Analysis notes — Section 9: Differences by Public Transport Use Frequency

Completion time by public transport use frequency

Participants with high public transport use frequency (n = 9) had a median completion time of 240 seconds, while participants with moderate use frequency (n = 7) had a median of 120 seconds. The dot plot shows considerable spread within both groups, particularly among the high-frequency users, whose completion times ranged from relatively short to the longest time observed in the dataset. The Mann-Whitney test gave U = 49.0, p = 0.063 (say what it is, with the n). The test cannot distinguish it from chance in groups this small — that is 'no statistically significant difference', which is honest. It is not 'no difference' and not 'the groups performed the same'. Say the study was too small to detect anything but a large effect. **suggests:** in this sample, high-frequency users actually tended to take longer rather than complete the task faster. However, the test could not distinguish this observed difference from chance at the conventional p < .05 level, and the two groups are small, therefore cannot establish that public transport use frequency is associated with completion speed.

The qualitative and observational data should be checked to explain why some high-frequency users took longer. One possibility to examine is whether more experienced users considered more alternatives, compared the prototype against their existing knowledge, or questioned the information more carefully rather than simply completing the task faster. The interviews should also help determine whether the longer times reflected difficulty or more deliberate decision-making.

Decision confidence by public transport use frequency

Both groups had a median decision-confidence rating of 4. High-frequency users (n = 9) and moderate-frequency users (n = 7) showed substantial overlap in their ratings, with scores in both groups ranging from 3 to 5. The Mann-Whitney test gave U = 37.5, p = .526. This bears on the question of whether participants' prior public transport familiarity may contribute to their confidence when making disruption-related travel decisions. The register is **cannot establish**: the current sample provides no clear quantitative evidence that decision confidence differed according to public transport use frequency. This should not be interpreted as evidence that the groups are identical; with only 9 and 7 participants, the study is too small to detect anything other than a relatively large difference.

The interview data should therefore be used to examine where participants said their confidence came from — for example, the prototype information, familiarity with the route, prior public transport experience, or other transport tools. This is particularly important because the earlier questionnaire results suggested that final decision confidence was generally positive even though the extent to which the prototype information itself increased confidence was more variable.

Read: there IS an observed difference (say what it is, with the n). The test cannot distinguish it from chance in groups this small — that is 'no statistically significant difference', which is honest. It is not 'no difference' and not 'the groups performed the same'. Say the study was too small to detect anything but a large effect.

10 Were your interviewees typical of the whole sample?

Your six qualitative participants sit inside the sixteen to twenty. If they were the most confident, or the fastest, or all in one age band, that bounds what the interviews can say — and it is a limitation to write down now, not discover in Week 8. Section 3 showed what the six were; this section checks whether they did what the others did — the same tasks, the same scores. This is